Nick Leung EE446 Assignment 1 Discussion

Problem 1

e. Yes, we can improve on parts b, c, and d by combining all three steps. My method is to take the trained student model, prune it, and quantize it to int8. The smallest model before part e was the int8 quantized model with a size of 5.80 KB. All other models were larger than 5.80 KB or were less accurate than 100%. My method achieved 100% accuracy with a size of 3.69 KB.

Problem 2

1. There are classification, transfer learning, object detection, regression, and anomaly detection blocks.
   Example models:
   - Classification: determine a movement type from IMU data
   - Transfer learning: A teacher model can be trained to recognize different animals, and a student model can be fine-tuned to recognize different species of an animal.
   - Object detection: A model can detect if a person is wearing a mask.
   - Regression: A model can predict how many times a sports team will win given historical data.
   - Anomaly detection: A model can use an autoencoder to detect when the CPU usage of a computer spikes.
2. Dense layers connect every input neuron to every output neuron. This helps to learn nonlinear relationships in the data. 1D convolution layers are used for processing time series data or any type of data along one dimension. 2D convolution layers learn features on data on 2 dimensions, such as images. Reshape layers convert 1D data into higher-dimensional data, which could help data move through different types of layers (convolutional to dense or vice versa). Flatten is a type of reshape layer that flattens data into 1 dimension. Dropout layers randomly cut connections in a network which can help reduce overfitting.
3. The only model compression option available in the free version is int8. float32 is unoptimized.
4. There are synthetic image generators and synthetic voice generators. Image generation requires a prompt describing the image to create and label to generate an image. Options for models are DALL-E 3 or the Flux Pro model. For voice generation, OpenAI’s text to speech model is used. A phrase and label are required. Synthetic data is important in machine learning applications because there could be gaps in the original dataset that the synthetic data needs to fill. One could generate images with unique features or include the voice for a missing keyword.